BLOCKCHAIN EXPERIMENT 9

Aim - To implement a Private Ethereum Blockchain using Geth.

Theory -

Ethereum is an open-source, decentralized blockchain platform that enables developers to build and deploy **smart contracts** and **decentralized applications (DApps)**. Unlike public blockchains such as the Ethereum mainnet, a **Private Ethereum Blockchain** is restricted to specific participants, making it ideal for **development, testing, and enterprise applications** where confidentiality, control, and performance are essential.

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through **JSON-RPC, command-line, or JavaScript consoles**.

Setting up a **Private Ethereum Network** involves creating a customized blockchain environment isolated from the public Ethereum network. It enables developers to test smart contracts, consensus mechanisms, and network behavior without incurring gas costs or depending on public validators.

Steps Involved:

1. **Choosing a Network ID:**
2. Each private blockchain must have a unique network ID to differentiate it from public or other private networks.
3. **Choosing a Consensus Algorithm:**
4. Ethereum supports different consensus mechanisms such as **Proof of Work (PoW)** and **Proof of Authority (PoA)**. In private networks, **Clique (PoA)** is often used for faster block confirmation.
5. **Creating a Genesis Block:**
6. The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.
7. **Initializing the Geth Database:**
8. The geth init command initializes the blockchain using the **genesis.json** file, preparing the data directory for node operation.
9. **Setting up Networking:**
10. Nodes communicate using **enodes** (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.
11. **Running the Member Nodes:**

12. Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.
13. **Running a Signer (in Clique):**
14. In the **Clique consensus**, designated **signers** validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

Outputs -

1. First install all the necessary dependencies like curl, git, build-essential.
   sudo apt install wget curl git build-essential -y

```
anushka@Anushka:~$ sudo apt update
[sudo] password for anushka:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1576 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1875 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [252 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1167 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.8 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [342 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2757 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1663 kB]
```

```
223 packages can be upgraded. Run 'apt list --upgradable' to see them.
anushka@Anushka:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  ethtool libllvm20 mesa-libgallium
The following packages have been kept back:
  libgl1-amber-dri libglapi-mesa
The following packages will be upgraded:
  apparmor apport apport-core-dump-handler apt apt-utils base-files binutils binutils-common binutils-x86-64-linux-gnu
  bsdextrautils bsdutils cloud-init coreutils curl dconf-gsettings-backend dconf-service dhcpcd-base dirmngr
  distro-info-data dmsetup dpkg eject fdisk gcc-14-base gir1.2-glib-2.0 gir1.2-packagekitglib-1.0 git git-man gnupg
  gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpgconf gpgsm gpgv gtk-update-icon-cache gzip iproute2
  iputils-ping keyboxd kmod krb5-locales landscape-client landscape-common libacl1 libapparmor1 libapt-pkg6.0t64
```

```
anushka@Anushka:~$ sudo apt install wget curl git build-essential -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.8).
curl set to manually installed.
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
The following packages were automatically installed and are no longer requir
```

2. Install geth using wget

cd /usr/local/bin && sudo wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e5 01b3b0.tar.gz

```
anushka@Anushka:/usr/local/bin$ cd /usr/local/bin
anushka@Anushka:/usr/local/bin$ sudo wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e501b3b
0.tar.gz
--2026-04-08 08:42:39--  https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.12.0-e501b3b0.tar.gz
Resolving gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)... 20.60.40.164
Connecting to gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)|20.60.40.164|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 26856444 (26M) [application/octet-stream]
Saving to: 'geth-linux-amd64-1.12.0-e501b3b0.tar.gz.1'

geth-linux-amd64-1.12.0-e501b 100%[===================================================>]  25.61M  2.13MB/s    in 12s

2026-04-08 08:42:52 (2.12 MB/s) - 'geth-linux-amd64-1.12.0-e501b3b0.tar.gz.1' saved [26856444/26856444]
```

3. Extract the tar file and move the geth into /usr/local/bin/geth path & check the geth version
   sudo tar -xvf geth-linux-amd64-1.12.0-e501b3b0.tar.gz
   sudo mv geth-linux-amd64-1.12.0-e501b3b0/geth /usr/local/bin/geth geth version

```
anushka@Anushka:/usr/local/bin$ sudo tar -xvf geth-linux-amd64-1.12.0-e501b3b0.tar.gz
geth-linux-amd64-1.12.0-e501b3b0/
geth-linux-amd64-1.12.0-e501b3b0/COPYING
geth-linux-amd64-1.12.0-e501b3b0/geth
anushka@Anushka:/usr/local/bin$ sudo mv geth-linux-amd64-1.12.0-e501b3b0/geth /usr/local/bin/geth
anushka@Anushka:/usr/local/bin$ sudo chmod +x /usr/local/bin/geth
anushka@Anushka:/usr/local/bin$ geth version
Geth
Version: 1.12.0-stable
Git Commit: e501b3b05db8e169f67dc78b7b59bc352b3c638d
Git Commit Date: 20230525
Architecture: amd64
Go Version: go1.20.3
Operating System: linux
GOPATH=
GOROOT=
anushka@Anushka:/usr/local/bin$
```

4. Create a folder named private_ethereum_setup and create another 2 directories naming node1 and node2 individually inside private_ethereum_setup directory.
   mkdir private_ethereum_setup
   cd private_ethereum_setup && mkdir node1 node2

```
anushka@Anushka:/usr/local/bin$ cd ~
mkdir private_ethereum_setup
cd private_ethereum_setup
mkdir node1 node2
```

5. Check the files
6. Create account for node1
   geth --datadir "node1" account new

```
anushka@Anushka:~/private_ethereum_setup$ geth --datadir node1 account new
INFO [04-08|08:49:08.105] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-08|08:49:08.107] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x64e4bf7f9f2A58C450B1Dcf91fBc83A9d375eB67
Path of the secret key file: node1/keystore/UTC--2026-04-08T08-49-22.431556294Z--64e4bf7f9f2a58c450b1dcf91fbc83a9d375eb67

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

7. Do the same for node2
   geth --datadir "node2" account new

```
anushka@Anushka:~/private_ethereum_setup$ geth --datadir node2 account new
INFO [04-08|08:50:23.705] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-08|08:50:23.708] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0xA0fB533b242f6A1F6F989bd444AC3666C559D290
Path of the secret key file: node2/keystore/UTC--2026-04-08T08-50-29.045373769Z--a0fb533b242f6a1f6f989bd444ac3666c559d290

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!
```

8. Create a file called genesis.json in the same directory and copy paste these contents

```
anushka@Anushka: ~/private
GNU nano 7.2                                genesis.json *
{
 "config": {
   "chainId": 12345,
   "homesteadBlock": 0,
   "eip150Block": 0,
   "eip155Block": 0,
   "eip158Block": 0,
   "byzantiumBlock": 0,
   "constantinopleBlock": 0,
   "petersburgBlock": 0,
   "istanbulBlock": 0,
   "berlinBlock": 0,
   "londonBlock": 0,
   "clique": {
     "period": 5,
     "epoch": 30000
   }
 },
 "difficulty": "1",
 "gasLimit": "8000000",
 "extradata": "0x0000000000000000000000000000000000000000000000000000000000000000e4bf7f9f2a58c450b1dcf91fbc83a9d375eb670000000000000
 "alloc": {
   "0x64e4bf7f9f2a58c450B1Dcf91fBc83A9d375eB67": {
     "balance": "1000000000000000000000"
   },
   "0xa0fb533b242f6a1f6f989bd444ac3666c559d290": {
     "balance": "1000000000000000000000"
   }
 }
}
```

Make changes in the alloc part replace the public keys of the account that you have generated in earlier steps.

9. Initializing geth for node1 using the genesis.json file
   geth init --datadir node1 genesis.json

```
anushka@Anushka:~/private_ethereum_setup$ geth init --datadir node1 genesis.json
INFO [04-08|08:58:59.587] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-08|08:58:59.590] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08|08:58:59.595] Set global gas cap                       cap=50,000,000
INFO [04-08|08:58:59.596] Initializing the KZG library             backend=gokzg
INFO [04-08|08:58:59.659] Using pebble as the backing database
INFO [04-08|08:58:59.659] Allocated cache and file handles         database=/home/anushka/private_ethereum_setup/node1/geth/chaindata
 cache=16.00MiB handles=16
INFO [04-08|08:58:59.726] Opened ancient database                  database=/home/anushka/private_ethereum_setup/node1/geth/chaindata
/ancient/chain readonly=false
INFO [04-08|08:58:59.727] Writing custom genesis block
INFO [04-08|08:58:59.727] Persisted trie from memory database      nodes=3 size=411.00B time="103.531µs" gcnodes=0 gcsize=0.00B gctim
e=0s livenodes=0 livesize=0.00B
INFO [04-08|08:58:59.752] Successfully wrote genesis state         database=chaindata hash=70f5d9..3a5e8f
INFO [04-08|08:58:59.752] Defaulting to pebble as the backing database
INFO [04-08|08:58:59.752] Allocated cache and file handles         database=/home/anushka/private_ethereum_setup/node1/geth/lightchai
ndata cache=16.00MiB handles=16
INFO [04-08|08:58:59.943] Opened ancient database                  database=/home/anushka/private_ethereum_setup/node1/geth/lightchai
ndata/ancient/chain readonly=false
INFO [04-08|08:58:59.943] Writing custom genesis block
INFO [04-08|08:58:59.947] Persisted trie from memory database      nodes=3 size=411.00B time="20.134µs"  gcnodes=0 gcsize=0.00B gctim
e=0s livenodes=0 livesize=0.00B
INFO [04-08|08:58:59.972] Successfully wrote genesis state         database=lightchaindata hash=70f5d9..3a5e8f
anushka@Anushka:~/private_ethereum_setup$
```

10. Do the same for node2
    geth init --datadir node2 genesis.json

```
anushka@Anushka:~/private_ethereum_setup$ geth init --datadir node2 genesis.json
INFO [04-08|08:59:56.827] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-08|08:59:56.828] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08|08:59:56.835] Set global gas cap                       cap=50,000,000
INFO [04-08|08:59:56.835] Initializing the KZG library             backend=gokzg
INFO [04-08|08:59:56.908] Using pebble as the backing database
INFO [04-08|08:59:56.908] Allocated cache and file handles         database=/home/anushka/private_ethereum_setup/node2/geth/chaindata
 cache=16.00MiB handles=16
INFO [04-08|08:59:57.037] Opened ancient database                  database=/home/anushka/private_ethereum_setup/node2/geth/chaindata
/ancient/chain readonly=false
INFO [04-08|08:59:57.037] Writing custom genesis block
INFO [04-08|08:59:57.037] Persisted trie from memory database      nodes=3 size=411.00B time="26.485µs" gcnodes=0 gcsize=0.00B gctime
=0s livenodes=0 livesize=0.00B
INFO [04-08|08:59:57.054] Successfully wrote genesis state         database=chaindata hash=70f5d9..3a5e8f
INFO [04-08|08:59:57.054] Defaulting to pebble as the backing database
INFO [04-08|08:59:57.055] Allocated cache and file handles         database=/home/anushka/private_ethereum_setup/node2/geth/lightchai
ndata cache=16.00MiB handles=16
INFO [04-08|08:59:57.340] Opened ancient database                  database=/home/anushka/private_ethereum_setup/node2/geth/lightchai
ndata/ancient/chain readonly=false
INFO [04-08|08:59:57.340] Writing custom genesis block
INFO [04-08|08:59:57.341] Persisted trie from memory database      nodes=3 size=411.00B time="32.068µs" gcnodes=0 gcsize=0.00B gctime
=0s livenodes=0 livesize=0.00B
INFO [04-08|08:59:57.361] Successfully wrote genesis state         database=lightchaindata hash=70f5d9..3a5e8f
```

11. Now we will install bootnode for configuring both the nodes.
sudo add-apt-repository -y ppa:ethereum/ethereum sudo apt-get update
sudo apt-get install bootnode

```
anushka@Anushka:~/private_ethereum_setup$ sudo add-apt-repository -y ppa:ethereum/ethereum
sudo apt update
sudo apt install bootnode
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu/
Suites: noble
Components: main
'
More info: https://launchpad.net/~ethereum/+archive/ubuntu/ethereum
Adding repository.
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:5 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble InRelease [18.1 kB]
Get:6 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main amd64 Packages [2584 B]
Get:7 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble/main Translation-en [772 B]
Fetched 21.4 kB in 2s (11.5 kB/s)
Reading package lists... Done
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:5 https://ppa.launchpadcontent.net/ethereum/ethereum/ubuntu noble InRelease
Reading package lists... Done
```

12. Create a key for the bootnode and save it to boot.key in the folder
bootnode -genkey boot.key

```
Setting up bootnode (1.14.13+build30700+noble) ...
anushka@Anushka:~/private_ethereum_setup$ bootnode -genkey boot.key
anushka@Anushka:~/private_ethereum_setup$ ls
boot.key  genesis.json  node1  node2
anushka@Anushka:~/private_ethereum_setup$
```

13. Copy this command and copy the enode://…. This will be required later
bootnode -nodekey boot.key -verbosity 9 -addr :30305

```
anushka@Anushka:~/private_ethereum_setup$ bootnode -nodekey boot.key -verbosity 9 -addr :30305
enode://9fe0a957b5d5feb9ae7c57c3f2605dc6345b7b09c671866eb6badabda7d60d03540a8dcff19d43649eaa4be1aab596747bfdc40f95ffb510654c30c78d352
772@127.0.0.1:0?discport=30305
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [04-08|09:02:44.072] New local node record                    seq=1,775,638,964,066 id=9f545fc70dede6ea ip="invalid IP" udp=0 tc
p=0
```

14. Now on another terminal we will start mining node1

```
geth --datadir node1 \
--port 30306 \
--bootnodes
"enode://9fe0a957b5d5feb9ae7c57c3f2605dc6345b7b09c671866eb6badabda7d60d03540a8dcff19d43649eaa4be1aab596747bfdc40f95ffb5
10654c30c78d352772@127.0.0.1:0?discport=30305" \
--networkid 123454321 \
--unlock 0x64e4bf7f9f2a58c450B1Dcf91fBc83A9d375eB67 \
--authrpc.port 8551 \
--miner.etherbase 0x64e4bf7f9f2a58c450B1Dcf91fBc83A9d375eB67 \
--mine
```

```
anushka@Anushka:~/private_ethereum_setup$ geth --datadir node1 \
--port 30306 \
--bootnodes "enode://9fe0a957b5d5feb9ae7c57c3f2605dc6345b7b09c671866eb6badabda7d60d03540a8dcff19d43649eaa4be1aab596747bfdc40f95ffb510
654c30c78d352772@127.0.0.1:0?discport=30305" \
--networkid 123454321 \
--unlock 0x64e4bf7f9f2a58c450B1Dcf91fBc83A9d375eB67 \
--authrpc.port 8551 \
--miner.etherbase 0x64e4bf7f9f2a58c450B1Dcf91fBc83A9d375eB67 \
--mine
INFO [04-08|09:10:59.847] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-08|09:10:59.849] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08|09:10:59.859] Set global gas cap                       cap=50,000,000
INFO [04-08|09:10:59.859] Initializing the KZG library             backend=gokzg
INFO [04-08|09:10:59.950] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-08|09:10:59.953] Using pebble as the backing database
INFO [04-08|09:10:59.954] Allocated cache and file handles         database=/home/anushka/private_ethereum_setup/node1/geth/chaindata
 cache=512.00MiB handles=524,288
INFO [04-08|09:11:00.065] Opened ancient database                  database=/home/anushka/private_ethereum_setup/node1/geth/chaindata
/ancient/chain readonly=false
INFO [04-08|09:11:00.083] Initialising Ethereum protocol           network=123,454,321 dbversion=<nil>
INFO [04-08|09:11:00.084]
INFO [04-08|09:11:00.084] ----------------------------------------------------------------------------------------------------
--------------------------------------------
INFO [04-08|09:11:00.084] Chain ID:  12345 (unknown)
INFO [04-08|09:11:00.084] Consensus: Clique (proof-of-authority)
```

```
ed=4.137ms
INFO [04-08|09:11:20.278] Looking for peers                        peercount=0 tried=0 static=0
INFO [04-08|09:11:25.003] Successfully sealed new block            number=5 sealhash=bd3084..4e3df4 hash=d6f402..8066a7 elapsed=4.998
s
INFO [04-08|09:11:25.003] "🔨 mined potential block"                number=5 hash=d6f402..8066a7
INFO [04-08|09:11:25.004] Commit new sealing work                  number=6 sealhash=970d49..0ea821 uncles=0 txs=0 gas=0 fees=0 elaps
ed=1.205ms
INFO [04-08|09:11:25.005] Commit new sealing work                  number=6 sealhash=970d49..0ea821 uncles=0 txs=0 gas=0 fees=0 elaps
ed=1.961ms
INFO [04-08|09:11:30.009] Successfully sealed new block            number=6 sealhash=970d49..0ea821 hash=5face4..a8453e elapsed=5.004
s
INFO [04-08|09:11:30.009] "🔨 mined potential block"                number=6 hash=5face4..a8453e
INFO [04-08|09:11:30.012] Commit new sealing work                  number=7 sealhash=c15eb5..a5cbb3 uncles=0 txs=0 gas=0 fees=0 elaps
ed=2.970ms
INFO [04-08|09:11:30.015] Commit new sealing work                  number=7 sealhash=c15eb5..a5cbb3 uncles=0 txs=0 gas=0 fees=0 elaps
ed=5.994ms
INFO [04-08|09:11:30.304] Looking for peers                        peercount=0 tried=0 static=0
INFO [04-08|09:11:35.002] Successfully sealed new block            number=7 sealhash=c15eb5..a5cbb3 hash=2cbcac..2c9177 elapsed=4.991
s
INFO [04-08|09:11:35.002] "🔨 mined potential block"                number=7 hash=2cbcac..2c9177
INFO [04-08|09:11:35.004] Commit new sealing work                  number=8 sealhash=115298..06b250 uncles=0 txs=0 gas=0 fees=0 elaps
ed=1.685ms
INFO [04-08|09:11:35.004] Commit new sealing work                  number=8 sealhash=115298..06b250 uncles=0 txs=0 gas=0 fees=0 elaps
ed=2.370ms
INFO [04-08|09:11:40.002] Successfully sealed new block            number=8 sealhash=115298..06b250 hash=b8b76b..4c44d6 elapsed=4.998
s
INFO [04-08|09:11:40.003] "🔗 block reached canonical chain"        number=1 hash=45b398..854ebd
INFO [04-08|09:11:40.003] "🔨 mined potential block"                number=8 hash=b8b76b..4c44d6
INFO [04-08|09:11:40.004] Commit new sealing work                  number=9 sealhash=865fff..467410 uncles=0 txs=0 gas=0 fees=0 elaps
ed=1.703ms
INFO [04-08|09:11:40.005] Commit new sealing work                  number=9 sealhash=865fff..467410 uncles=0 txs=0 gas=0 fees=0 elaps
ed=2.751ms
INFO [04-08|09:11:40.343] Looking for peers                        peercount=0 tried=0 static=0
```

15. Now do same for node2

geth --datadir node2 \
--port 30307 \
--bootnodes "enode://9fe0a957b5d5feb9ae7c57c3f2605dc6345b7b09c671866eb6badabda7d60d03540a8dcff19d43649eaa4be1aab596747bfdc40f95ffb510654c30c78d352772@127.0.0.1:0?discport=30305" \
--networkid 123454321 \
--unlock 0xa0fb533b242f6a1f6f989bd444ac3666c559d290 \
--authrpc.port 8552

```
PS C:\Users\anush>  wsl -d Ubuntu
anushka@Anushka:/mnt/c/Users/anush$  cd ~/private_ethereum_setup
anushka@Anushka:~/private_ethereum_setup$ geth --datadir node2 \
--port 30307 \
--bootnodes "enode://9fe0a957b5d5feb9ae7c57c3f2605dc6345b7b09c671866eb6badabda7d60d03540a8dcff19d43649eaa4be1aab596747bfdc40f95ffb510
654c30c78d352772@127.0.0.1:0?discport=30305" \
--networkid 123454321 \
--unlock 0xa0fb533b242f6a1f6f989bd444ac3666c559d290 \
--authrpc.port 8552
INFO [04-08|09:13:36.678] Maximum peer count                       ETH=50 LES=0 total=50
INFO [04-08|09:13:36.681] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-08|09:13:36.698] Set global gas cap                       cap=50,000,000
INFO [04-08|09:13:36.699] Initializing the KZG library             backend=gokzg
INFO [04-08|09:13:36.790] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-08|09:13:36.791] Using pebble as the backing database
INFO [04-08|09:13:36.791] Allocated cache and file handles         database=/home/anushka/private_ethereum_setup/node2/geth/chaindata
 cache=512.00MiB handles=524,288
INFO [04-08|09:13:36.893] Opened ancient database                  database=/home/anushka/private_ethereum_setup/node2/geth/chaindata
/ancient/chain readonly=false
INFO [04-08|09:13:36.903] Initialising Ethereum protocol           network=123,454,321 dbversion=<nil>
INFO [04-08|09:13:36.904]
INFO [04-08|09:13:36.904] ----------------------------------------------------------------------------------------------------
-----------------------------------------------
INFO [04-08|09:13:36.904] Chain ID:  12345 (unknown)
INFO [04-08|09:13:36.904] Consensus: Clique (proof-of-authority)
INFO [04-08|09:13:36.904]
INFO [04-08|09:13:36.904] Pre-Merge hard forks (block based):
INFO [04-08|09:13:36.904]  - Homestead:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-
upgrades/mainnet-upgrades/homestead.md)
INFO [04-08|09:13:36.904]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/blob/master/network-
upgrades/mainnet-upgrades/tangerine-whistle.md)
INFO [04-08|09:13:36.904]  - Spurious Dragon/1 (EIP 155): #0        (https://github.com/ethereum/execution-specs/blob/master/network-
```

```
Unlocking account 0xa0fb533b242f6a1f6f989bd444ac3666c559d290 | Attempt 1/3
Password:
INFO [04-08|09:13:42.846] Unlocked account                         address=0xA0fB533b242f6A1F6F989bd444AC3666C559D290
INFO [04-08|09:13:46.997] Block synchronisation started
INFO [04-08|09:13:47.031] Looking for peers                        peercount=1 tried=1 static=0
INFO [04-08|09:13:47.031] Syncing: state download in progress      synced=30.53% state=196.00B accounts=1@196.00B slots=0@0.00B codes
=0@0.00B eta=40.950ms
INFO [04-08|09:13:47.067] Syncing: state download in progress      synced=100.00% state=392.00B accounts=2@392.00B slots=0@0.00B code
s=0@0.00B eta=-86ns
INFO [04-08|09:13:47.067] Syncing: state healing in progress       accounts=0@0.00B   slots=0@0.00B codes=0@0.00B nodes=0@0.00B pendi
ng=0
INFO [04-08|09:13:47.096] Imported new chain segment               number=33 hash=d3b770..7f762e blocks=33 txs=0 mgas=0.000 elapsed=1
8.716ms    mgasps=0.000 dirty=0.00B
INFO [04-08|09:13:47.097] Syncing: chain download in progress      synced=+Inf%   chain=18.00B headers=33@6.00B bodies=33@6.00B recei
pts=33@6.00B eta=-20.539ms
INFO [04-08|09:13:47.097] Snap sync complete, auto disabling
INFO [04-08|09:13:50.006] Imported new chain segment               number=34 hash=3e044e..9796fe blocks=1  txs=0 mgas=0.000 elapsed="
814.537µs" mgasps=0.000 dirty=0.00B
INFO [04-08|09:13:55.004] Imported new chain segment               number=35 hash=d1e301..37a29a blocks=1  txs=0 mgas=0.000 elapsed="
379.503µs" mgasps=0.000 dirty=0.00B
INFO [04-08|09:13:57.057] Looking for peers                        peercount=1 tried=0 static=0
INFO [04-08|09:14:00.005] Imported new chain segment               number=36 hash=737c59..17d373 blocks=1  txs=0 mgas=0.000 elapsed="
476.952µs" mgasps=0.000 dirty=0.00B
INFO [04-08|09:14:05.003] Imported new chain segment               number=37 hash=17656c..d7a208 blocks=1  txs=0 mgas=0.000 elapsed="
258.143µs" mgasps=0.000 dirty=0.00B
INFO [04-08|09:14:07.085] Looking for peers                        peercount=1 tried=0 static=0
INFO [04-08|09:14:10.005] Imported new chain segment               number=38 hash=b98b20..d18831 blocks=1  txs=0 mgas=0.000 elapsed="
355.379µs" mgasps=0.000 dirty=0.00B
```

```
TRACE[04-08|09:16:53.668] << FINDNODE/v4                          id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:53.673] >> NEIGHBORS/v4                         id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:53.832] << FINDNODE/v4                          id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:53.833] >> NEIGHBORS/v4                         id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:54.170] << FINDNODE/v4                          id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:54.171] >> NEIGHBORS/v4                         id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:54.333] << FINDNODE/v4                          id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:54.334] >> NEIGHBORS/v4                         id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:54.672] << FINDNODE/v4                          id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:54.673] >> NEIGHBORS/v4                         id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:54.835] << FINDNODE/v4                          id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:54.836] >> NEIGHBORS/v4                         id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:55.175] << FINDNODE/v4                          id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:55.175] >> NEIGHBORS/v4                         id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:55.337] << FINDNODE/v4                          id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:55.338] >> NEIGHBORS/v4                         id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:55.678] << FINDNODE/v4                          id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:55.678] >> NEIGHBORS/v4                         id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:55.840] << FINDNODE/v4                          id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:55.841] >> NEIGHBORS/v4                         id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:56.181] << FINDNODE/v4                          id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:56.182] >> NEIGHBORS/v4                         id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:56.342] << FINDNODE/v4                          id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:56.343] >> NEIGHBORS/v4                         id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:56.682] << FINDNODE/v4                          id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:56.695] >> NEIGHBORS/v4                         id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:56.760] >> PING/v4                              id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:56.764] << PONG/v4                              id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:56.765] >> ENRREQUEST/v4                        id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:56.768] << ENRRESPONSE/v4                       id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
DEBUG[04-08|09:16:56.768] Node revalidated                        b=15 id=c1b186052bf325f2 checks=64 q=slow
TRACE[04-08|09:16:56.843] << FINDNODE/v4                          id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:56.843] >> NEIGHBORS/v4                         id=c1b186052bf325f2 addr=127.0.0.1:30306 err=<nil>
TRACE[04-08|09:16:57.185] << FINDNODE/v4                          id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
TRACE[04-08|09:16:57.186] >> NEIGHBORS/v4                         id=c6342ef16b4b9dd5 addr=127.0.0.1:30307 err=<nil>
```

16. Now we will interact with the blockchain using javascript console
    geth attach node1/geth.ipc

```
anushka@Anushka:~/private_ethereum_setup$ geth attach node1/geth.ipc
Welcome to the Geth JavaScript console!

instance: Geth/v1.12.0-stable-e501b3b0/linux-amd64/go1.20.3
coinbase: 0x64e4bf7f9f2a58c450b1dcf91fbc83a9d375eb67
at block: 57 (Wed Apr 08 2026 09:15:45 GMT+0000 (UTC))
 datadir: /home/anushka/private_ethereum_setup/node1
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> eth.accounts
["0x64e4bf7f9f2a58c450b1dcf91fbc83a9d375eb67"]
> eth.getBalance(eth.accounts[0])
```

17. Interacting with the blockchain

```
To exit, press ctrl-d or type exit
> eth.accounts
["0x64e4bf7f9f2a58c450b1dcf91fbc83a9d375eb67"]
> eth.getBalance(eth.accounts[0])
1e+21
> eth.getBalance(eth.accounts[0])
1e+21
> eth.sendTransaction({
......   from: "0x64e4bf7f9f2a58c450B1Dcf91fBc83A9d375eB67",
......   to: "0xa0fb533b242f6a1f6f989bd444ac3666c559d290",
......   value: web3.toWei(1, "ether")
...... })
"0x3fd7f8bf87201aae426f5378bc6d1ddaba3e22a4f2083f0723f68bae92437600"
```

18. Sending transactions between accounts

```
> eth.sendTransaction({
......    from: "0x64e4bf7f9f2a58c450B1Dcf91fBc83A9d375eB67",
......    to: "0xa0fb533b242f6a1f6f989bd444ac3666c559d290",
......    value: web3.toWei(1, "ether")
...... })
"0x3fd7f8bf87201aae426f5378bc6d1ddaba3e22a4f2083f0723f68bae92437600"
>
```

19. Analyzing transactions

```
> eth.getTransaction("0x3fd7f8bf87201aae426f5378bc6d1ddaba3e22a4f2083f0723f68bae92437600")
{
  accessList: [],
  blockHash: "0x7bfdffce3e1208fa57b68c998d62829a761072332511a05c12a415bce8fd1aa4",
  blockNumber: 83,
  chainId: "0x3039",
  from: "0x64e4bf7f9f2a58c450b1dcf91fbc83a9d375eb67",
  gas: 21000,
  gasPrice: 1000015373,
  hash: "0x3fd7f8bf87201aae426f5378bc6d1ddaba3e22a4f2083f0723f68bae92437600",
  input: "0x",
  maxFeePerGas: 1000035138,
  maxPriorityFeePerGas: 1000000000,
  nonce: 0,
  r: "0xa6ccf82ff1d7a904a27d0c2134d385215cc6c8995537794098e95b35a4d51653",
  s: "0x4bfee035f55f294df298f870839cbc5adca4324c1a9e9584660d738138381caa",
  to: "0xa0fb533b242f6a1f6f989bd444ac3666c559d290",
  transactionIndex: 0,
  type: "0x2",
  v: "0x0",
  value: 1000000000000000000
}
>
```

```
> eth.getBlock(83)
{
  baseFeePerGas: 15373,
  difficulty: 2,
  extraData: "0xd883010c00846765746888676f312e32302e33856c696e75780000000000000177fbb039e07259023d5b90a21ac0982b10bd26a8cbf5d4e6c645
c2c0e5971cc3eca0bdd166acc5a2d941ba92833230f0ea3066244befd33beefe94ab28be9dc00",
  gasLimit: 8674973,
  gasUsed: 21000,
  hash: "0x7bfdffce3e1208fa57b68c998d62829a761072332511a05c12a415bce8fd1aa4",
  logsBloom: "0x00000000000000000000
```